MINING DEALER BEST PRACTICE

Lifting Tool to Support Large Planetaries

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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1.0 Introduction

The assembly of the large planetaries (992) involves the manipulation of heavy gears. These gears create a danger for the technician during the lifting and placement of the gear and the process is time consuming.

A lifting tool was designed to support the gear during installation and alignment with the pin bores. This tool reduces the safety risk, parts damage, and decreases the assembly time.

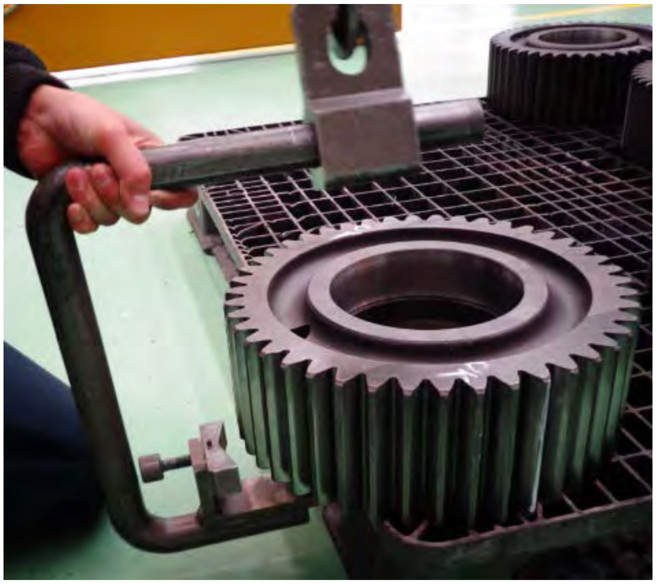
2.0 Best Practice Description

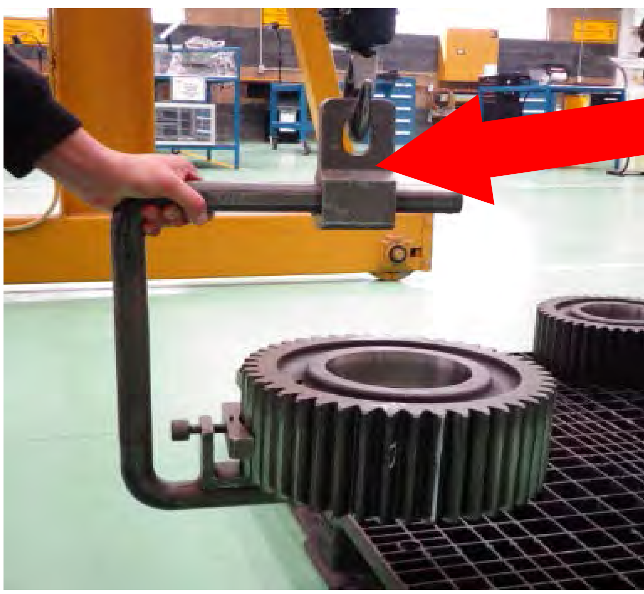
- Easy transportation
- Easy handling
- Easy storage
- Reliant and adjustable for different gears sizes.
- High adhesive paint to eliminate contamination from the painted support surface
- Minimum maintenance

3.0 Implementation Steps

Install the planetary carrier on the support stand (see illustrations). Support stand has adjustable carrier supports (6 inches in height) that have lip stops that prevent movement of the planetary carrier.

The majority of large planetary gears (992) have one orifice. This orifice is used with the lifting tool to help secure the planetary gear to the tool.

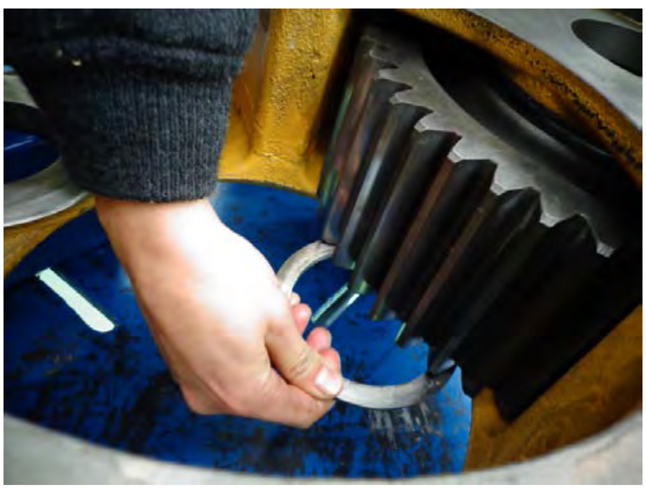


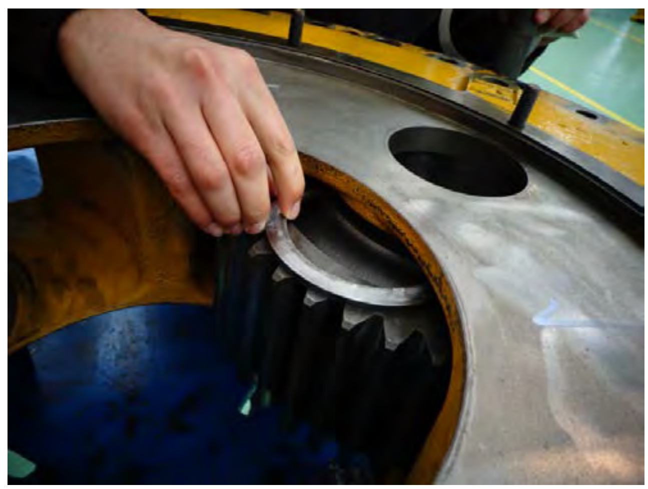


Adjustable for balancing the weight of the different planetary gears

With the gear secured on the lifting tool. The repair area crane is used to lift and manipulate the gear while installing the gear in to the carrier.

When the gear is in the planetary carrier the thrust washers or spacers are installed before removing the lifting tool.





When the gear and the thrust washers or spacers are installed, release the tension on the adjusting screw and remove the lifting tool.

Once the lifting tool is removed you can align and install the carrier gear pin.

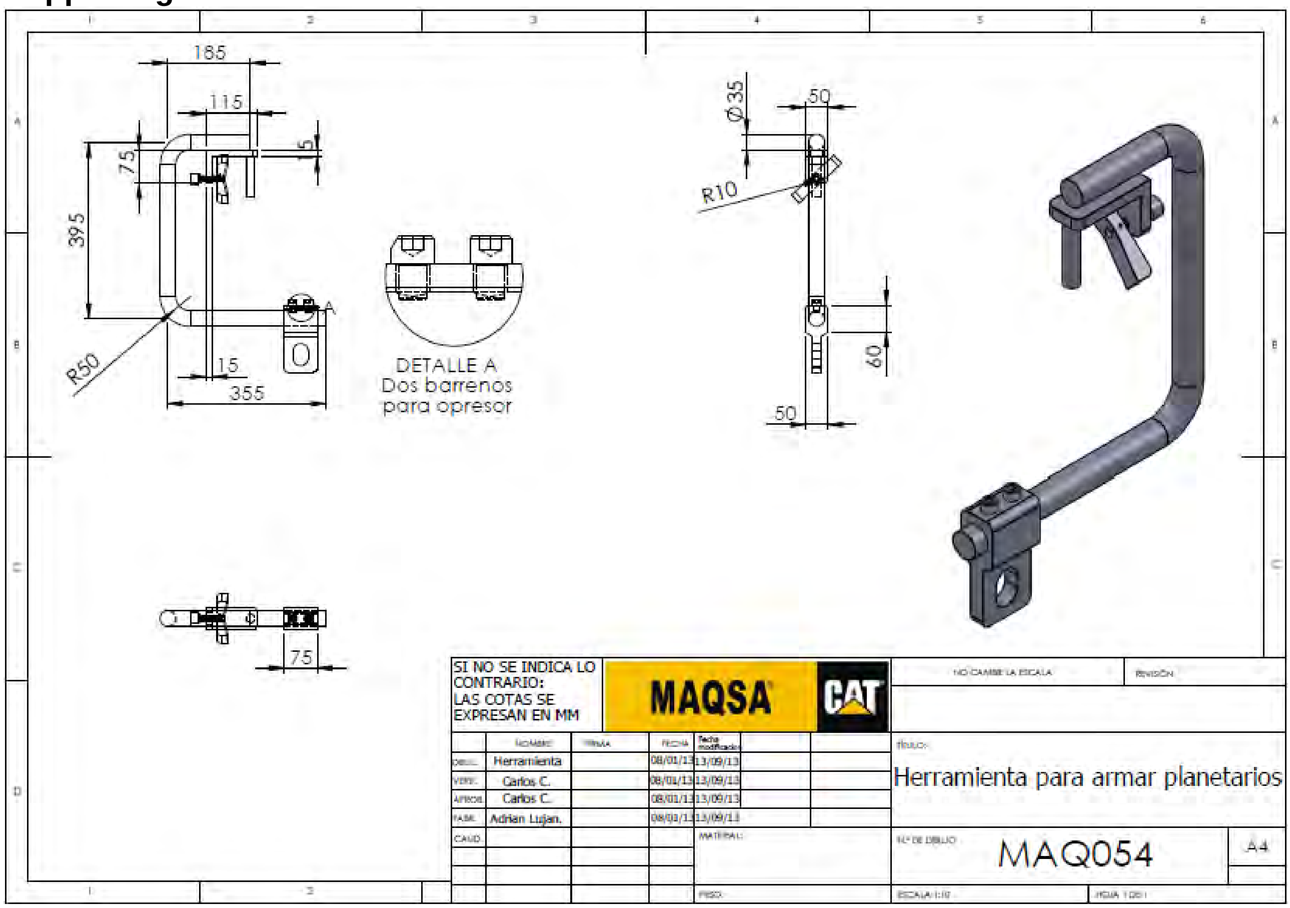
4.0 Benefits

-The installation of the large planetary gears is safer for the technician and the assembly is faster.
-With this tool the technician does not have to do any lifting of planetary gears.
-This tool can also be used when a hydraulic press is needed to install the planetary gears that require pressing the pin in to the carrier.

5.0 Resources Required

Fabrication shop
8-10 labor hours.
Material cost, less than $1000.00

6.0 Supporting Attachments / References



7.0 Related Best Practices

8.0 Acknowledgements

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